



Marine Biology

海洋生物学



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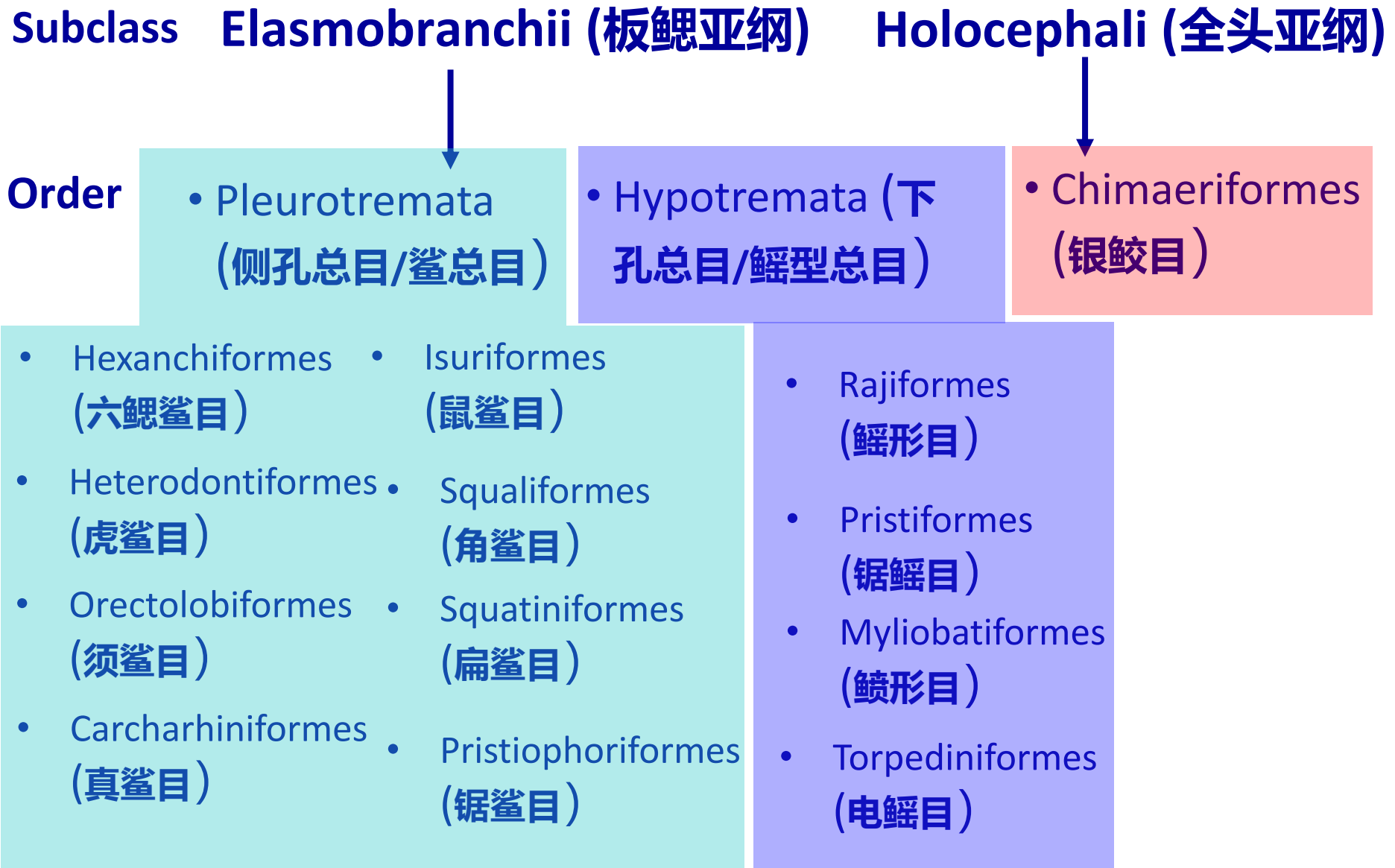
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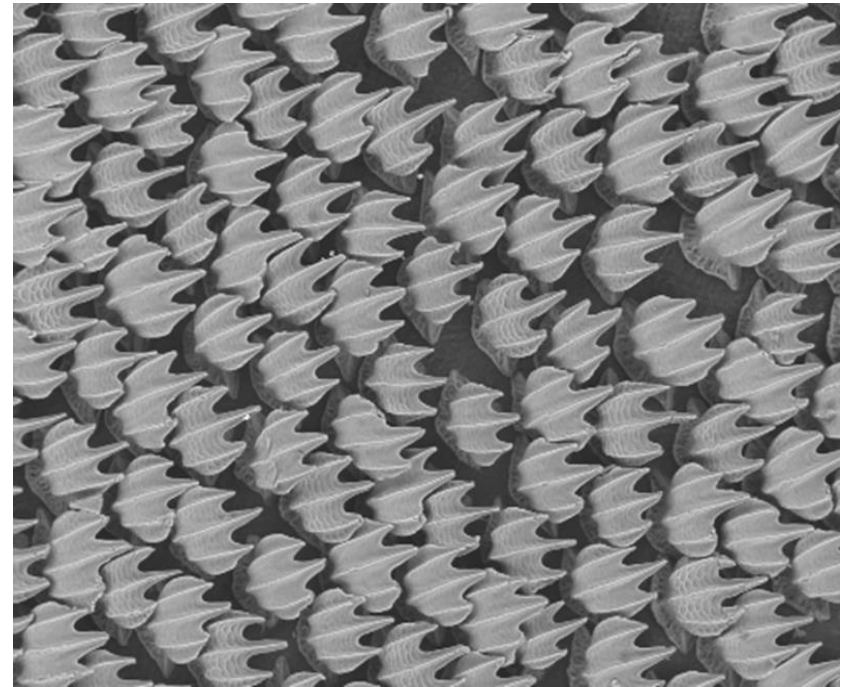
More Advanced Groups of Fish

- Fishes in the Chondrichthyes (**软骨鱼类**) and Osteichthyes (**硬骨鱼类**)
- General Characteristics/advancements in these groups:
 - Highly efficient **gills**
 - **Scales** cover the body
 - Paired **fins**
 - A wide variety of **jaw** and feeding types
 - **Lateral line** and other sensory organs
 - Streamlined body

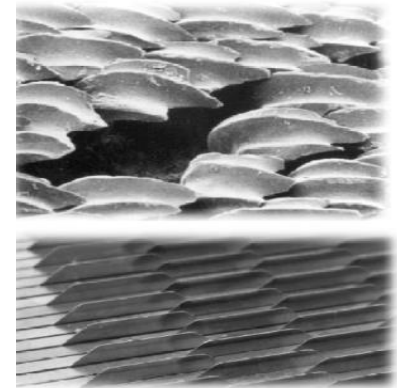
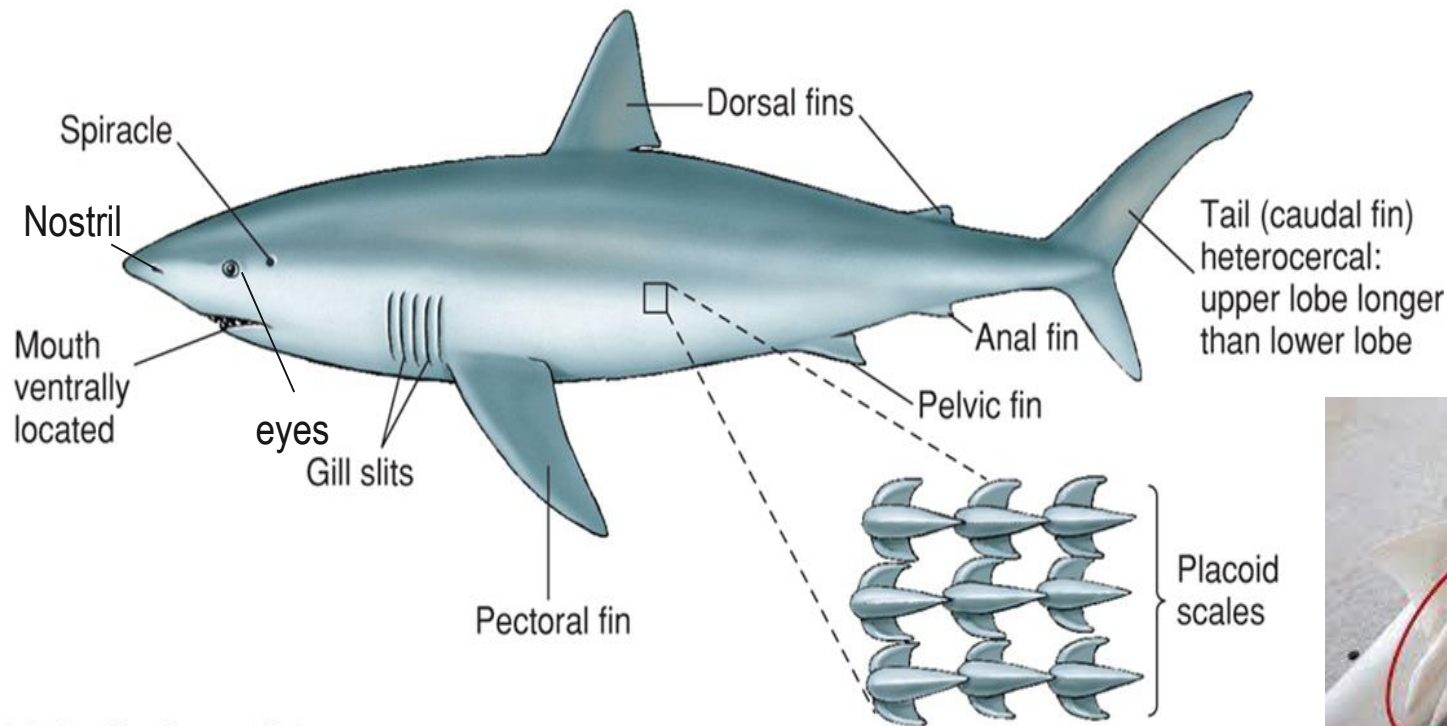
3. Chondrichthyes: 2 subclasses and 13 orders



- About 1000 species, including sharks (鲨鱼), skates (鳐鱼), rays (魟鱼) and ratfishes (银鲛)
- Skeleton of cartilage (<https://microbenotes.com/bone-vs-cartilage/>)
- **Lateral line** as sensor organ
- Movable jaws with well-developed teeth
- **Placoid scales** (盾鳞) and paired fins
- 5-7 gill slits open directly into the water in most species

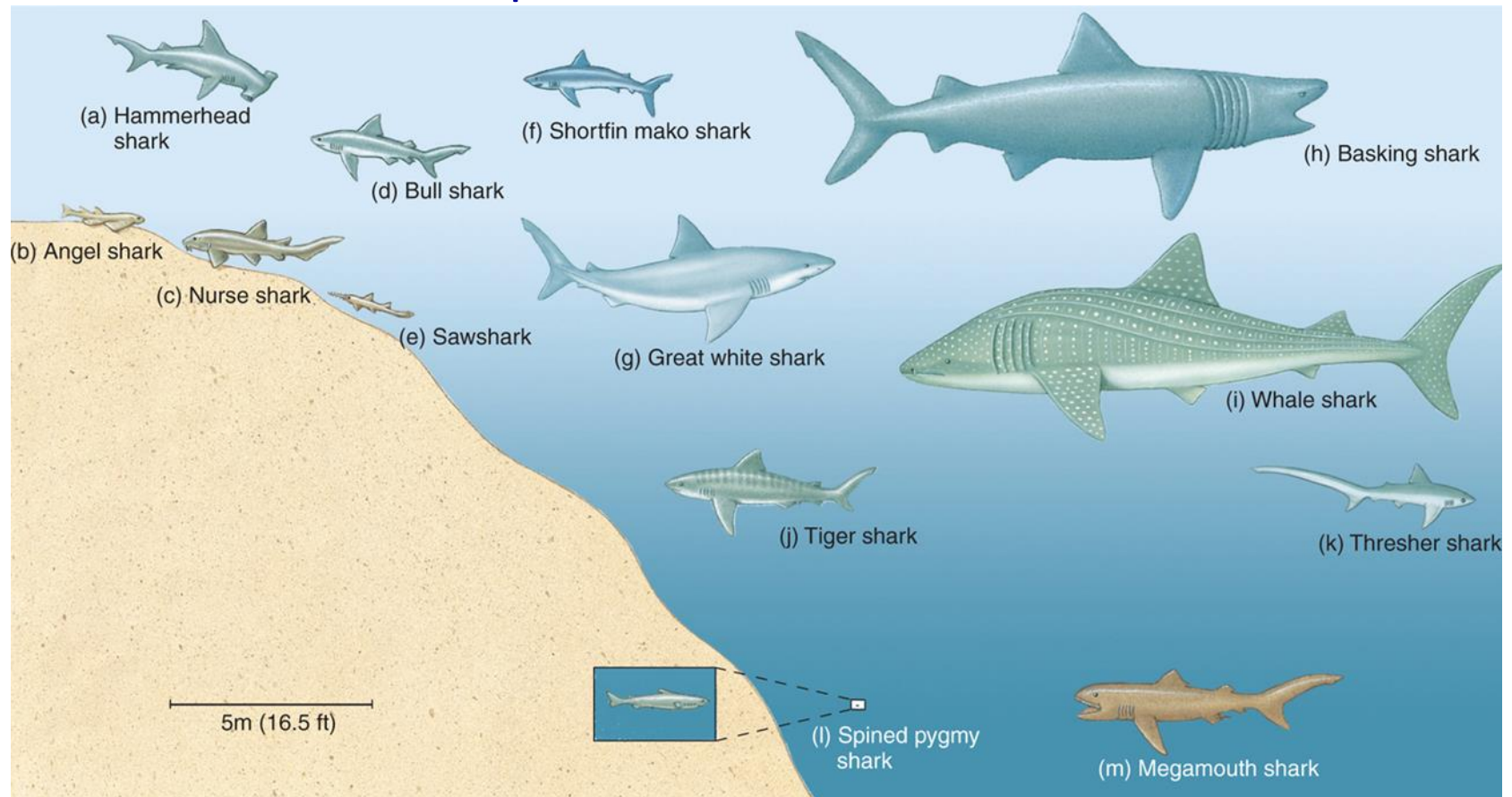


- **Spiracles (喷气孔)** in many species (openings on head used to directly bring water in for respiration without opening the mouth)
- Males in most species have projections of the pelvic fin called **claspers (卷须)** that are used in copulation

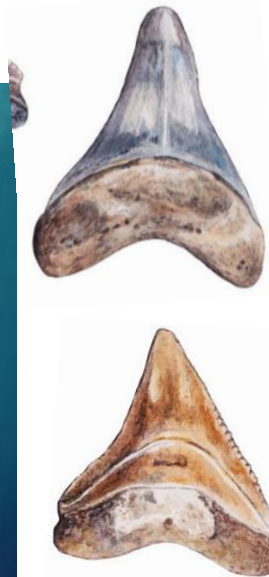


3.1 Sharks

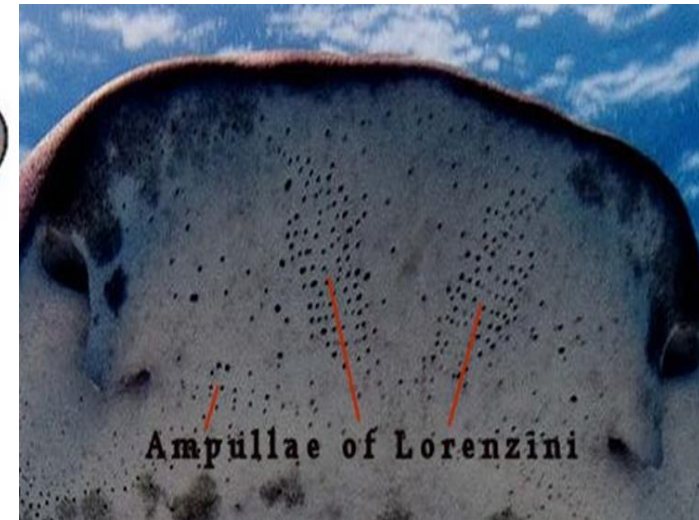
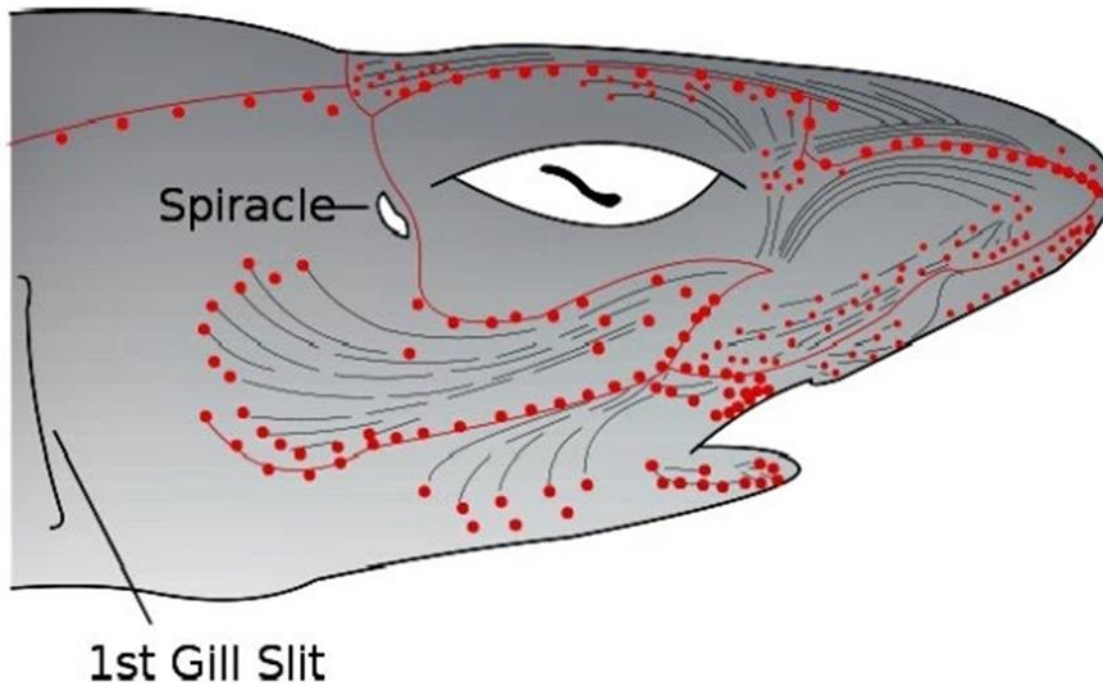
- About 350 species currently exist, many are carnivorous, some planktonic
- Primarily marine, but do travel up the mouths of rivers into freshwater in some species



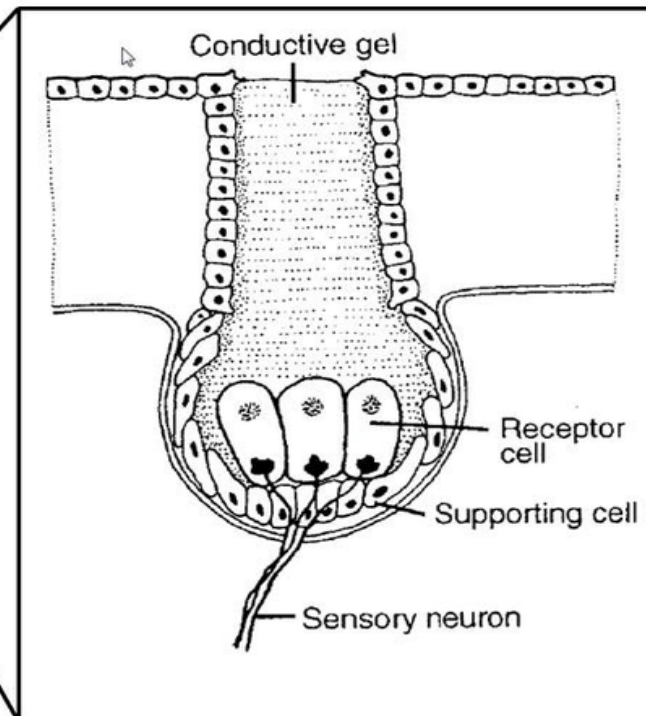
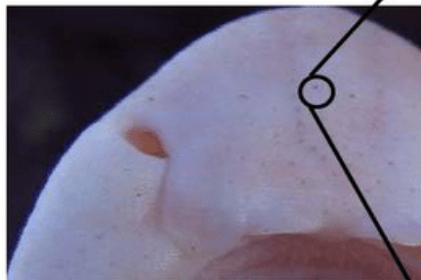
- ❑ Most species possess 2 dorsal fins
- ❑ Powerful jaw has rows of numerous sharp, often triangular teeth which are embedded in a tough, fibrous membrane covering the jaw
- ❑ A lost or broken tooth is replaced by another, which slowly shifts forward from the row behind it as if on a conveyor belt

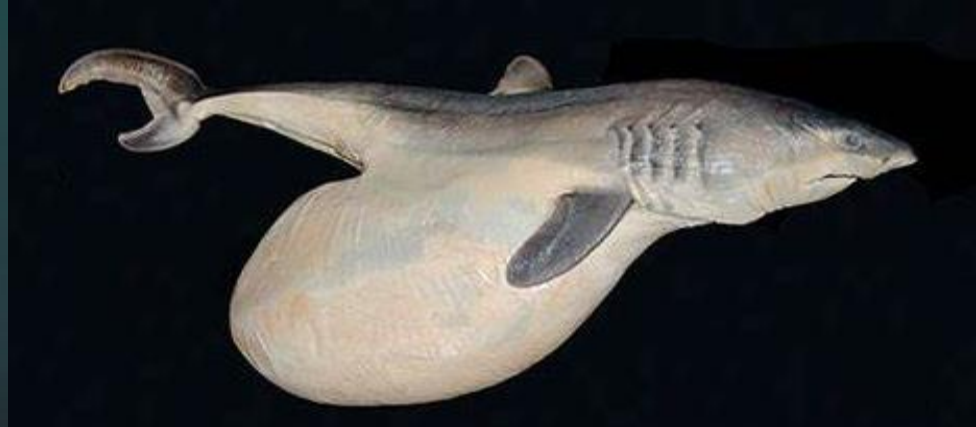


- In addition to the lateral line, sharks have a sensory organ called the **Ampullae of Lorenzini (罗伦氏器)**; this organ is used to detect electrical currents when sharks are close to an object, allows sharks to detect living prey even it is hidden from their view if they can get within a few feet of the organism



- Each ampullae's only external feature are the pits that line the faces and nostril of sharks and stingrays
- Pits are filled with an electro-receptive crystalline gel which is so sensitive and can detect voltage as low as to 15 billionths (十亿分之) of a volt per square centimeter
- Once electricity is sensed, the gel triggers a synapse (突触) response in the shark's brain based on the total information the external electrical current provides

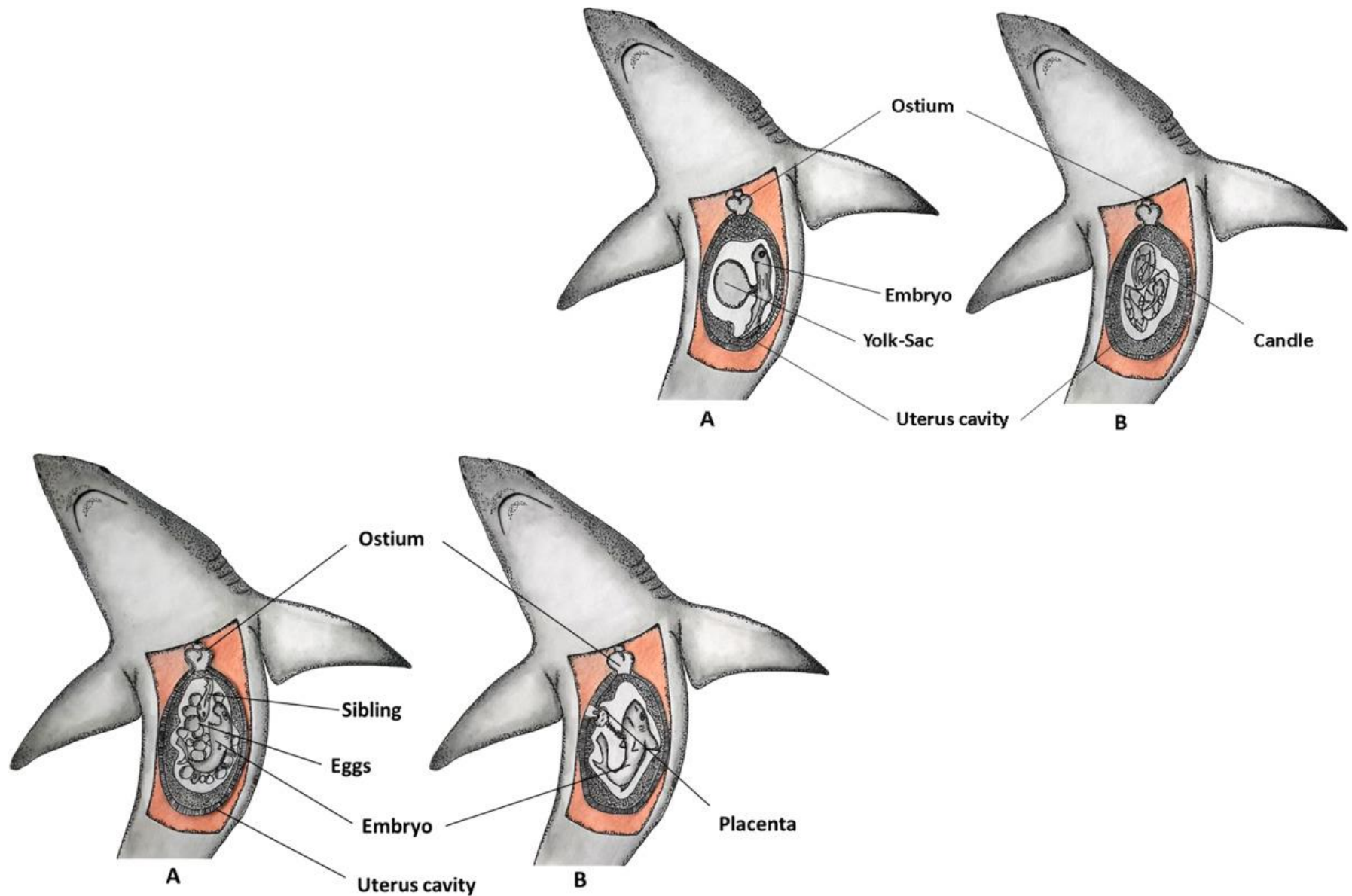




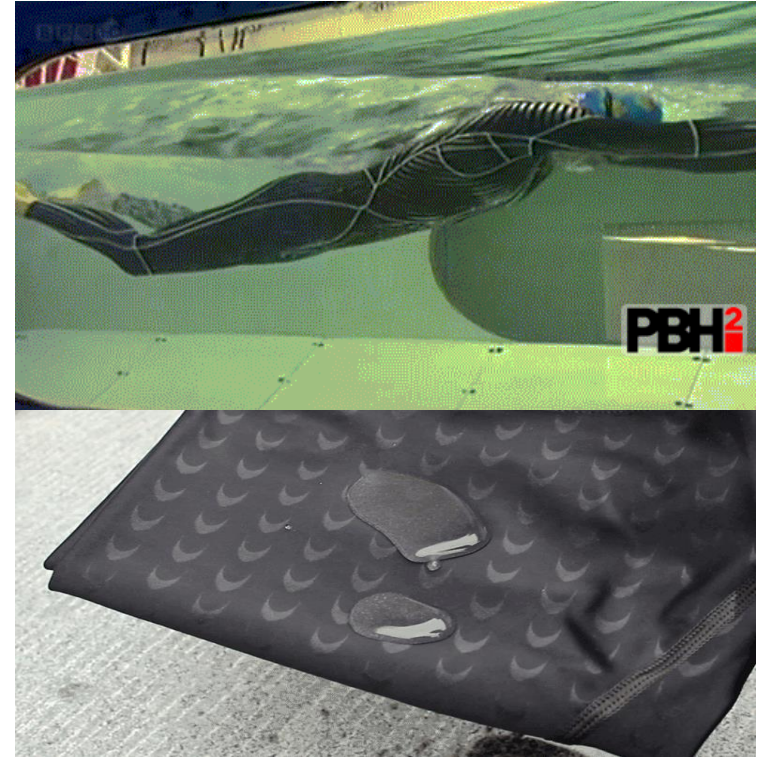
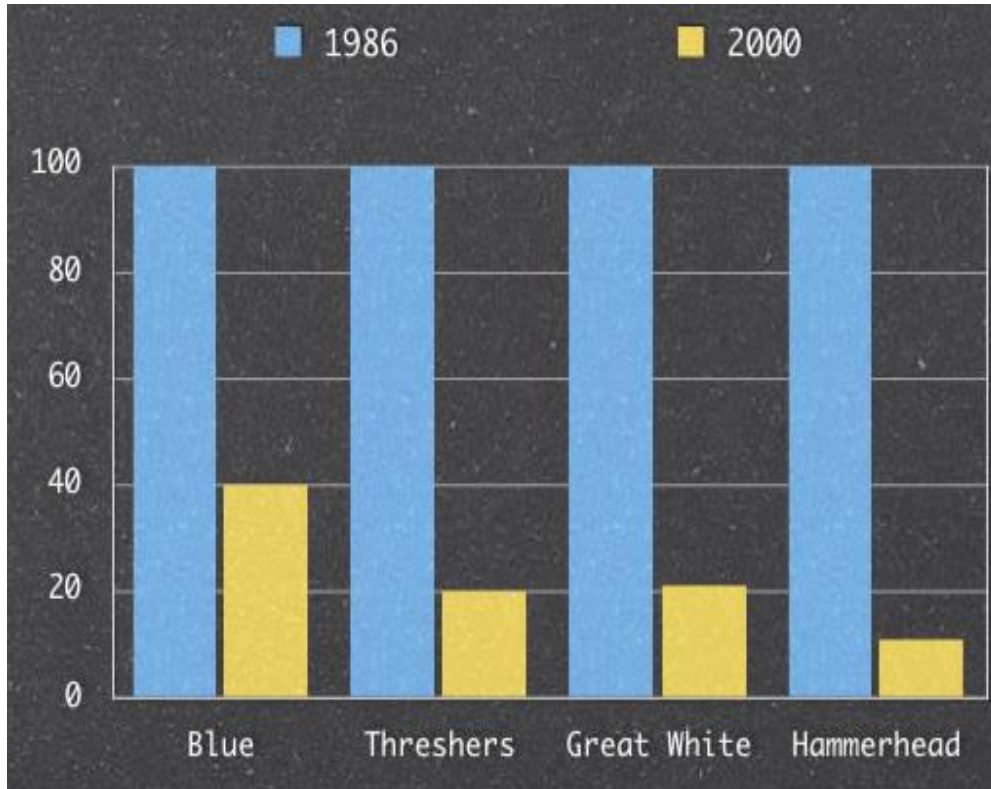
- Reproduction in sharks is varied:
 - some have live birth (known as vivipary, **胎生**);
 - some lay egg cases (ovipary, **卵生**);
 - some have an internalized egg that hatches while still in the female before the young are expelled from the female's reproductive system (ovovivipary, **卵胎生**)

Comparative Reproduction

Cynthia A. Awruch, in [Encyclopedia of Reproduction \(Second Edition\)](#), 2018



- The population numbers in many species of sharks are declining due to harvesting sharks for meat, oil, skin and fins



❑ Many species have included in the international protecting list

Sharks of the ocean

Selected shark species that are vulnerable, threatened with extinction or endangered

Illustrations not to scale

Angel

Squatina squatina
Critically Endangered



Maximum length 1.8 m

An ambush predator that buries itself in sand in the temperate waters of the European and North African continental shelves

Broadfin

Lamiopsis temminckii
Endangered



1.7 m

Once thought common, now rare. Found inshore on the continental shelves of the Indian Ocean and West Pacific. Harmless to humans, eaten by humans

Great Hammerhead

Sphyrna mokarran
Endangered



6 m

Their wide-set eyes give them a broad visual range. An organ inside their head allows them to detect the electric field of buried stingrays

Great White

Carcharodon carcharias
Vulnerable



6 m

Species most commonly found to attack humans. New borns immediately try to swim away from their mother, which might try to eat them

Greenland

Somniosus microcephalus
Near Threatened



6.5 m

Native to the cold, deep waters of the North Atlantic. Believed to live for at least 272 years, possibly as long as 500

Pondicherry

Carcharhinus homiodon
Critically endangered
(Possibly even extinct: IUCN)



2 m

Also known as the long-nosed shark. Last recorded in India in 1979. Little known about their size or breeding habits

Scalloped Hammerhead

Sphyrna lewini
Endangered



3.3 m

They can dive up to 1,000 m but are mostly found near the surface. Small individuals known to migrate in large schools, but larger adults tend to be solitary

Tiger

Galeocerdo cuvier
Near Threatened

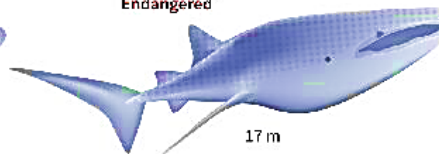


5.5 m

Named for distinct black stripes found on juveniles. Second only to great whites in attacking people. Stomach contents have included plastics, metals, tire scraps, almost anything dumped in the sea

Whale

Rhincodon typus
Endangered



17 m

Biggest fish in the world. The gentle giants swim with their mouths open to swallow up floating plankton or small fish that happen to be passing by

Zebra

Stegostoma fasciatum
Endangered



2.5 m

Shocked scientists in 2017 when a captive female hatched eggs with three viable pups, despite not having contact with a male for at least three years, thereby switching from sexual to asexual reproduction



FACTS ABOUT SHARKS

- Cartilaginous skeletons, not bones.
- Cold-blooded; body temp matches surroundings.
- Sharp, replaceable teeth for tearing prey.
- They have sensitive electroreceptors.
- Excellent sense of smell for hunting.
- Streamlined bodies for fast swimming.
- Multiple species, diverse shapes and sizes.
- Powerful jaws with rows of teeth.
- Some species can jump out of water.
- Unique dorsal fin shapes for identification.

- Skin with tiny tooth-like structures.
- Gills extract oxygen from water.
- Most species are solitary hunters.
- Some sharks migrate thousands of miles.
- Top predators in marine food chains.
- Many species give birth to live young.
- Slow reproductive rates; vulnerable to overfishing.
- Highly adapted to their environments.
- Known for occasional attacks on humans.
- Play vital roles in ocean ecosystems.

3.2 Rays

- ❑ Rays always have **live birth- vivipary**
- ❑ Pectoral fins (胸鳍) are expanded into “wings” and entire body is dorsoventrally flattened
- ❑ **Gill slits (5 pairs)**, mouth are on the underside of the body
- ❑ Spiracles are located on the top of the body ; most have long whip-like tails

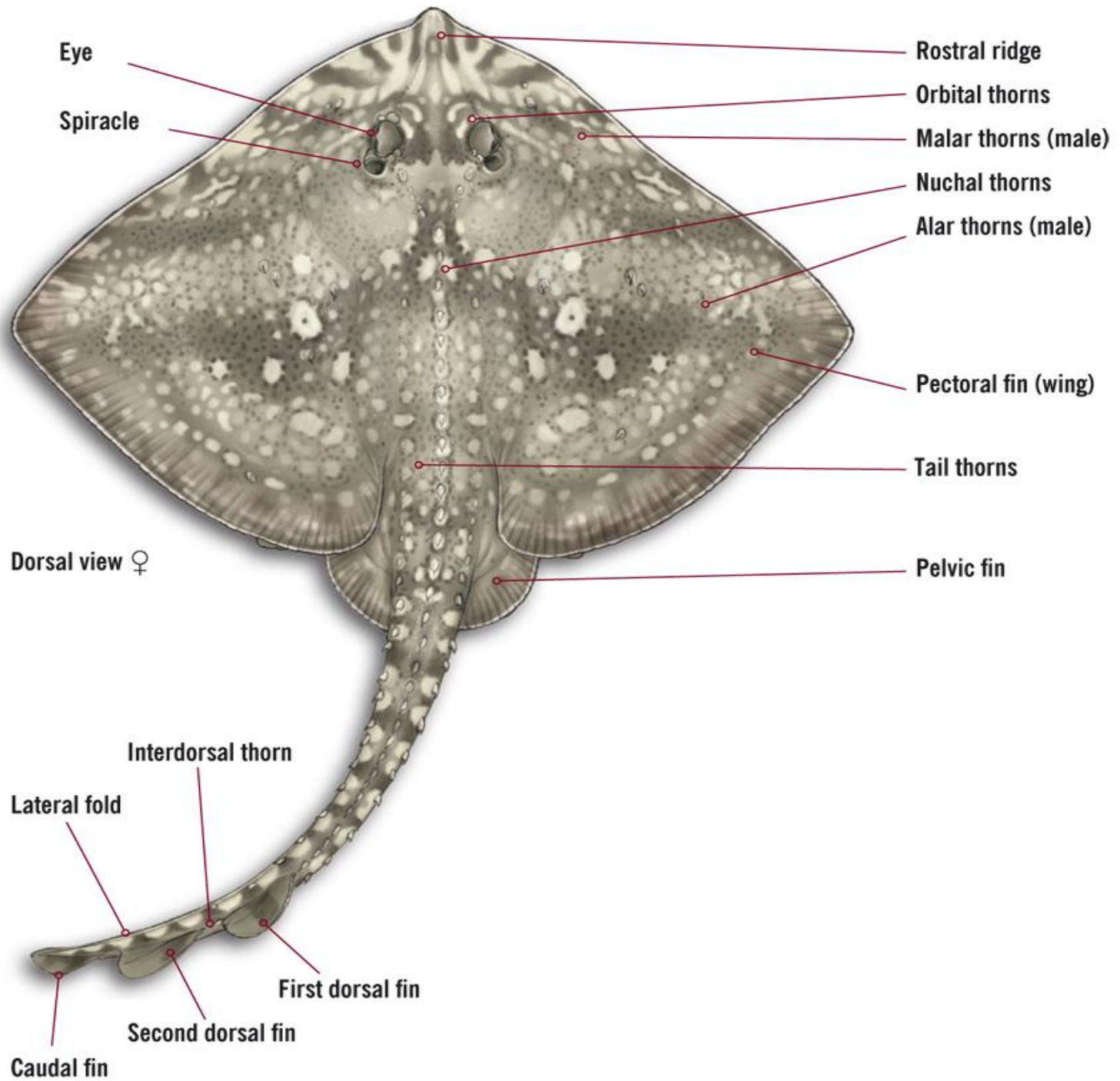


- ❑ In sting rays, there is a spine at the base of the tail with an associated poison gland, venom from this gland is delivered to other organisms
- ❑ Electric rays have organs that produce electricity on either side of head
- ❑ Rays spend much of their time on the bottom (demersal) partially covered in sand



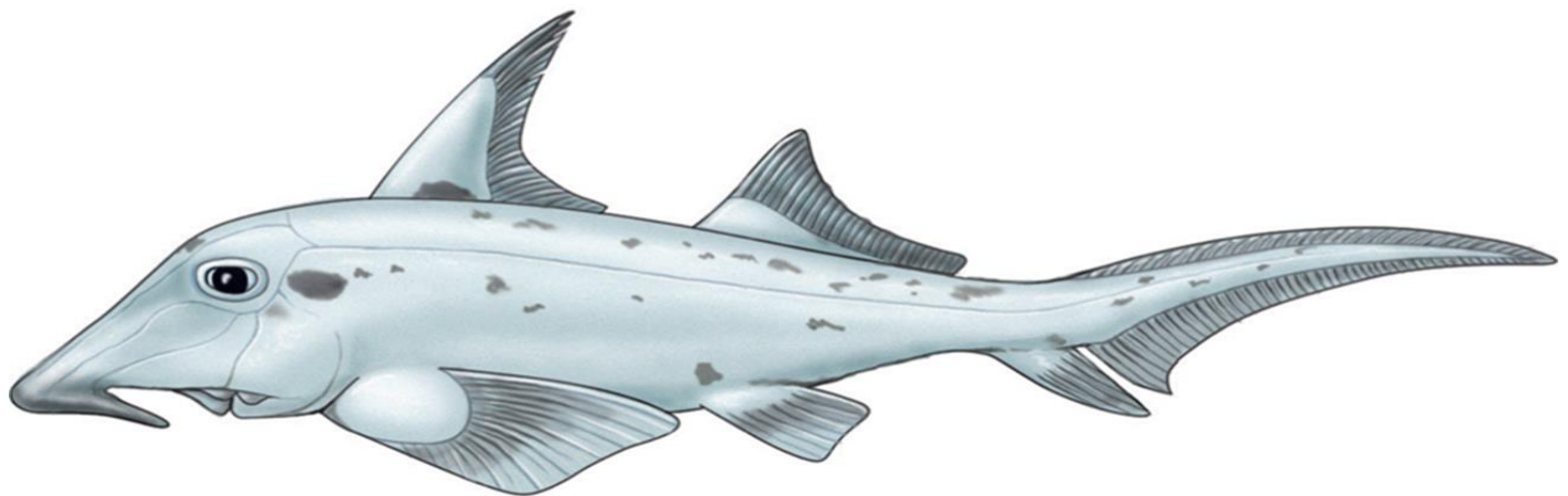
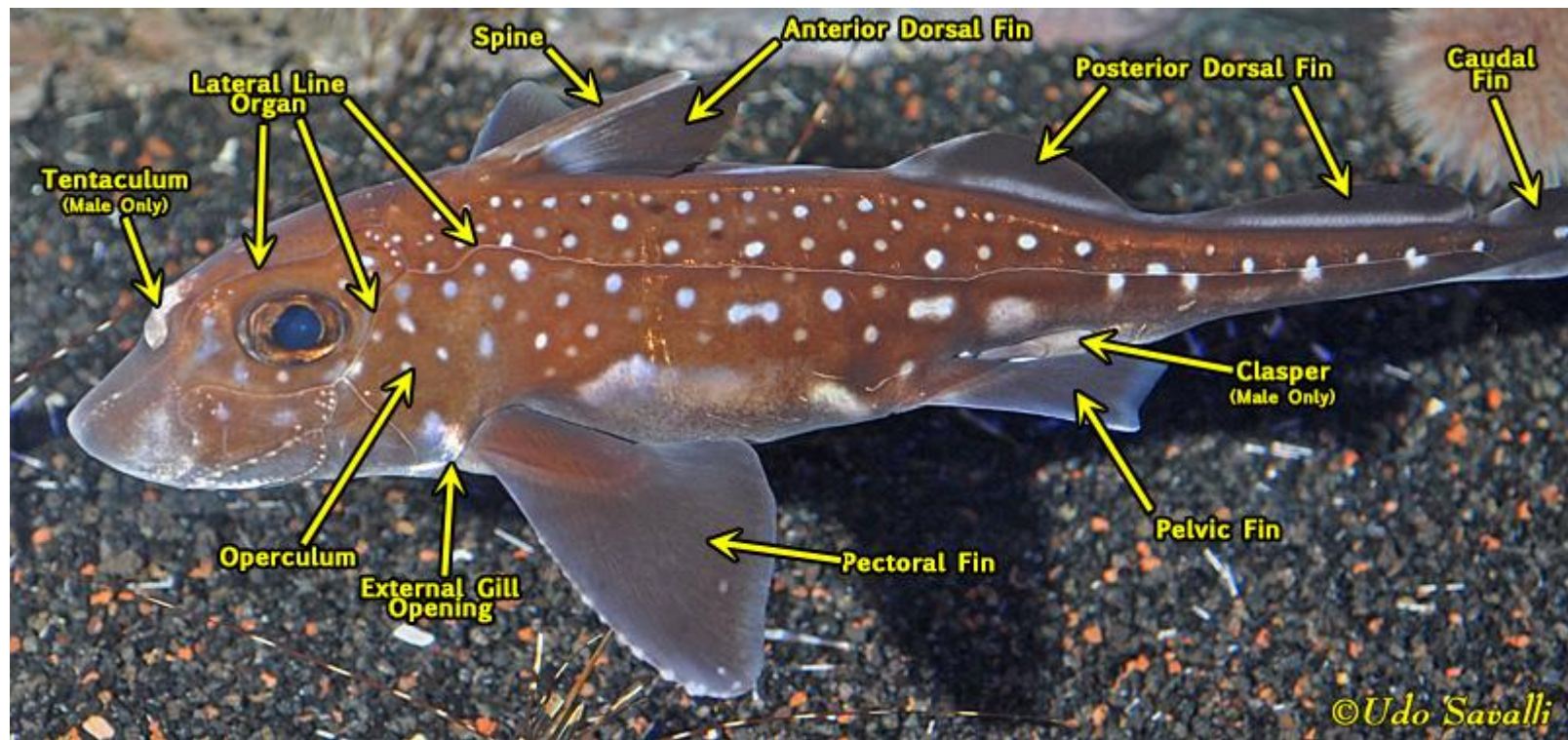
3.3 Skates

- ❑ **Like rays**, skates are dorsoventrally flattened with pectoral fins modified into wings
- ❑ They are also demersal and feed on mollusks and arthropods primarily
- ❑ **Unlike rays**, skates have a fleshy tail and **no spine** on the tail
- ❑ **Also unlike rays**, skates always lay egg cases:
 - ✓ After fertilization, the female lays egg cases with an embryo inside
 - ✓ The embryo develops within the protection of the egg case for weeks to months



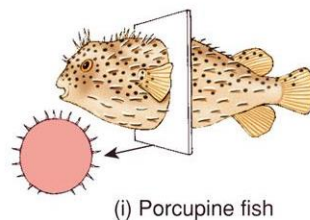
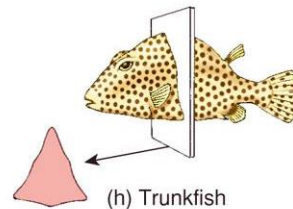
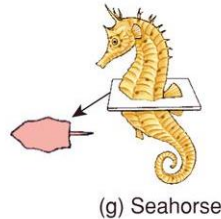
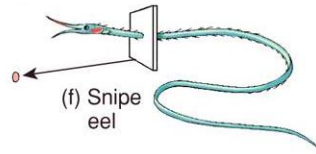
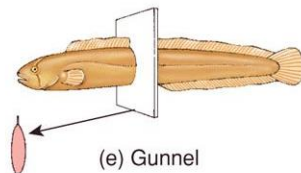
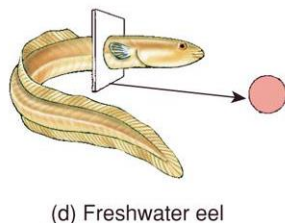
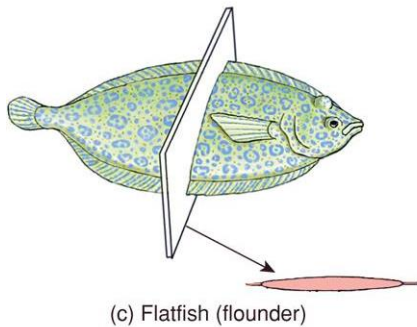
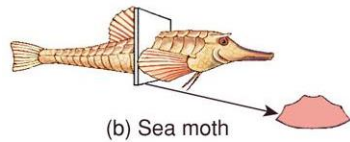
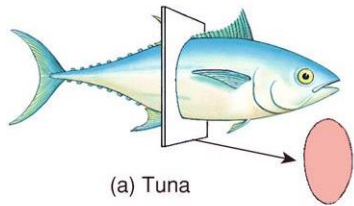
3.4 Ratfishes (Class of Holocephali, 全头亚纲)

- Only 1 class, about 30 species, mostly are **deep water** inhabitant
- **Four pairs of gill slits** is covered by a flap of skin (very **unusual** in the chondrichthyes), only one pair of gill slit hole appeared externally
- Feed on crustaceans and molluscs on the bottom primarily
- **Heterocercal (不等鳍的) tail** like in sharks
- **Unlike** others in this group, they also have **fin rays** (tiny support rods) in the fins
- Smooth skin without scale, placoid scales only appeared in larvae



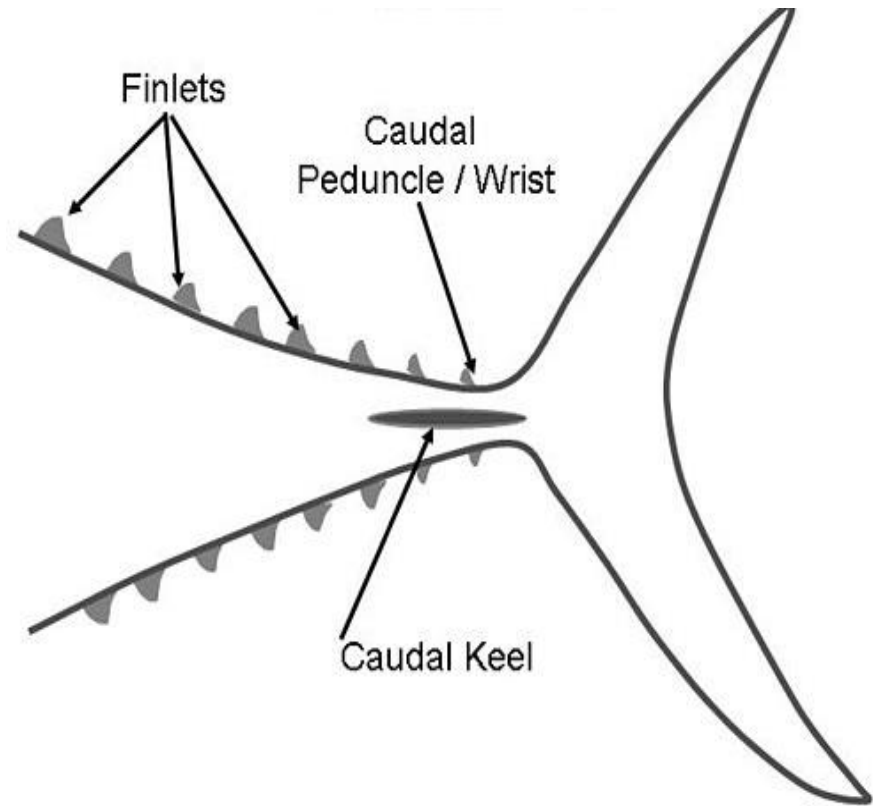
Biology of fishes

1. Body Shape: directly related to its lifestyle



- (a) Streamlined in **tunas** (金枪鱼)
- (b) Ventrally flattened in bottom-dwellers as in **sea moth** (海蛾鱼)
- (c) Laterally flattened in bottom-dwellers as in **flounder** (比目鱼)
- (d) **Eel-like** body for living among vegetation or rocks
- (e) Ribbon-shaped in **gunnel** (鰻鱼)
- (f) Thread-like in **snipe eel** (线鳗)
- (g) Elongated body on a vertical plane for slow swimmer, e.g. **seahorse**
- (h) Triangular as in **trunkfish** (硬鳞鱼)
- (i) Round as in the **porcupine fish** (刺鲀)

- Fast swimmers have a **streamlined** body shape to help move through the water
- There is an area of the body called the **caudal peduncle** (尾茎, area just before the tail) is very thin – this allows all the muscles to concentrate in this area allowing for greater thrust of the caudal (tail) fin (this means FAST swimming capabilities)



- **Demersal fishes** are **dorsoventrally flattened** for live on bottom
- Distinctly **elongated bodies** for live in narrow spaces in rocks or coral reefs

- **Laterally compressed bodies** are good for leisurely swimming around coral reefs, kelp beds, or rocky reefs, but **are still efficient enough to allow for bursts of speed to escape from enemies or capture food**

Example: Angelfish (Order of Perciformes, Family of Pomacanthidae, 鲈形目刺盖鱼科) do not inhabit the open waters of the ocean, but like to inhabit coral reefs, oyster reefs and other similar environments. Its body is not streamlined and the fins are feather-like for lots of flexibility. This flexibility allows for greater control and fit to live in a coral reef environment (crevices, etc.)



- Body shape can be useful for camouflage

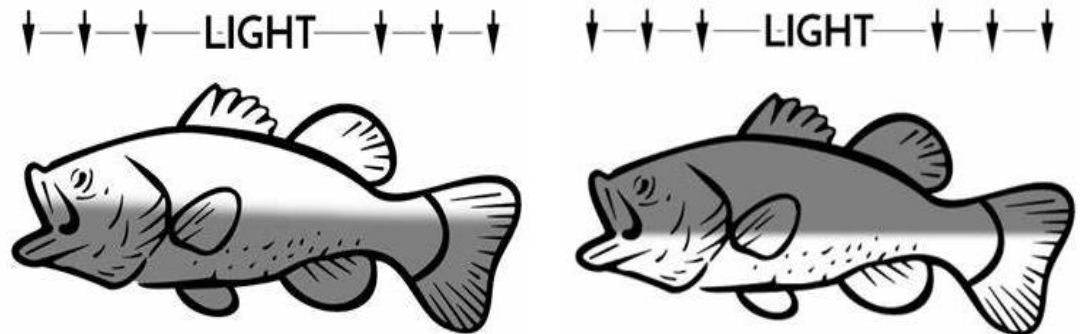
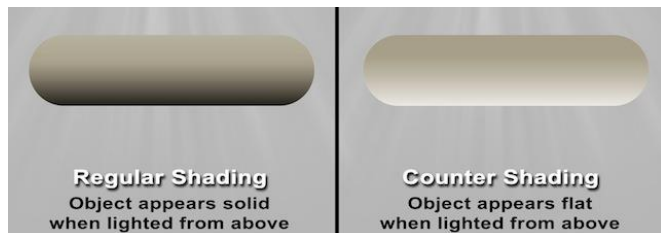
Example: Fish like the toadfish (蟾鱼目, Family of Bathachoidiforms) actually look like rocks or “scenery” and thus can go undetected by predators or prey



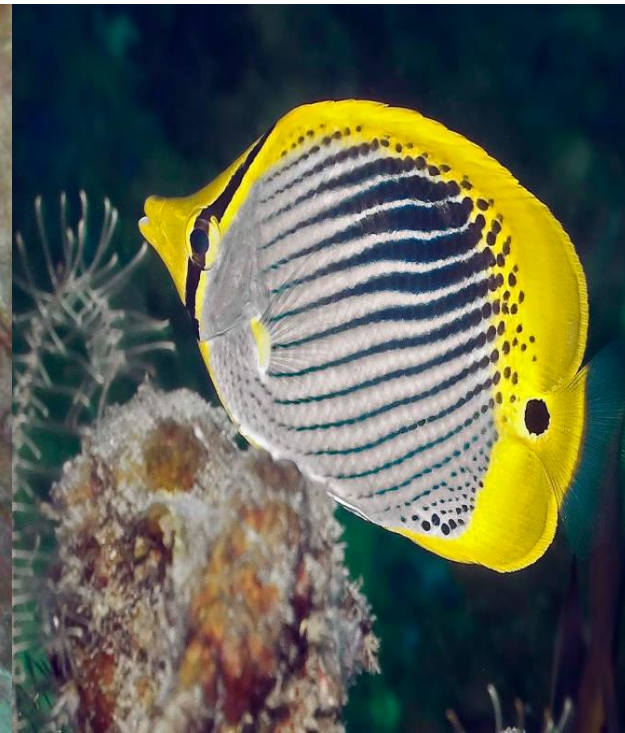
2. Coloration

1) Countershading (反隐蔽) is seen in all fish species

- The ventral (belly) area of the fish is lighter than the dorsal area of the fish
- Allows the fish to “blend in” with the environment
- If a fish is seen from above, the darker coloration of the dorsal area blends in with the darker color of the ocean bottom
- If the fish is seen from below, the lighter coloration of the ventral surface blends in with the lighter coloration of the ocean surface

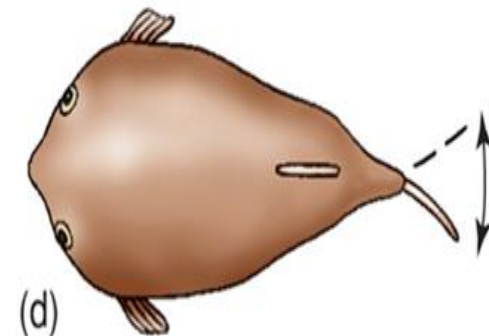
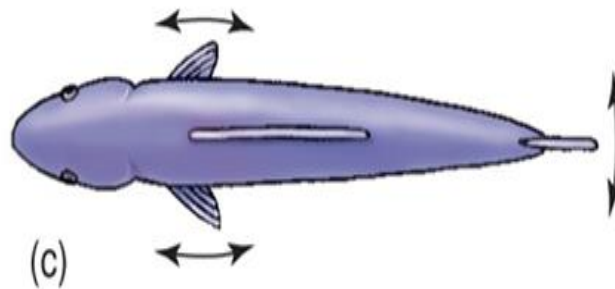
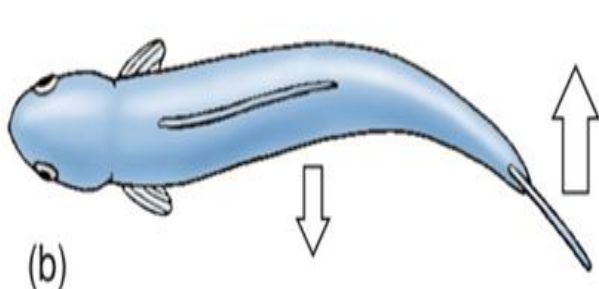
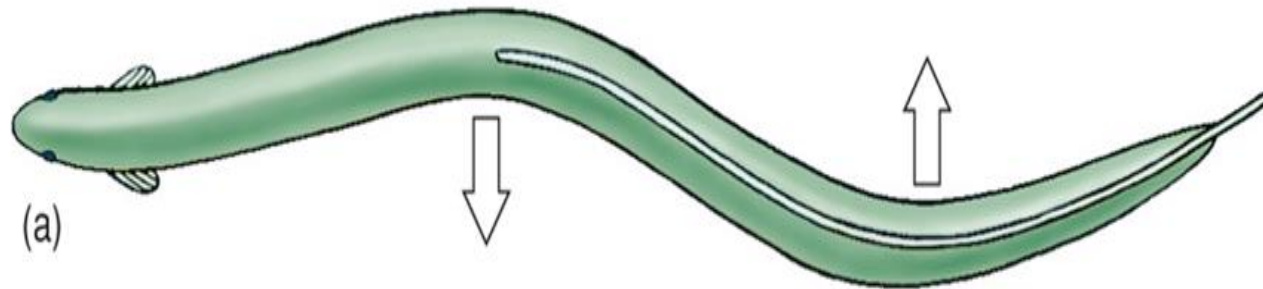
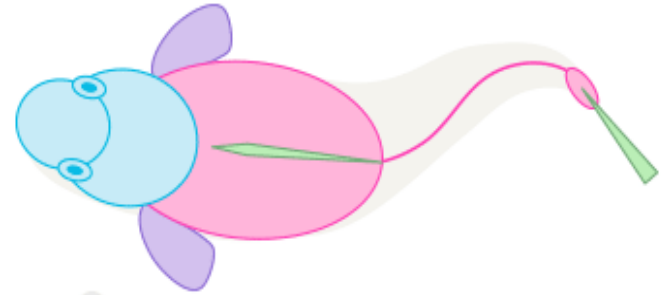


- 2) Slower swimming fish often have bars or stripes that help break up the silhouette (轮廓) of a fish (a form of **disruptive coloration**)
- 3) Fish with circular patterns on or near the caudal fin for **camouflage**
- 4) Some fishes may also use color to advertise their bad taste or poisonous nature – this is known as **warning coloration**

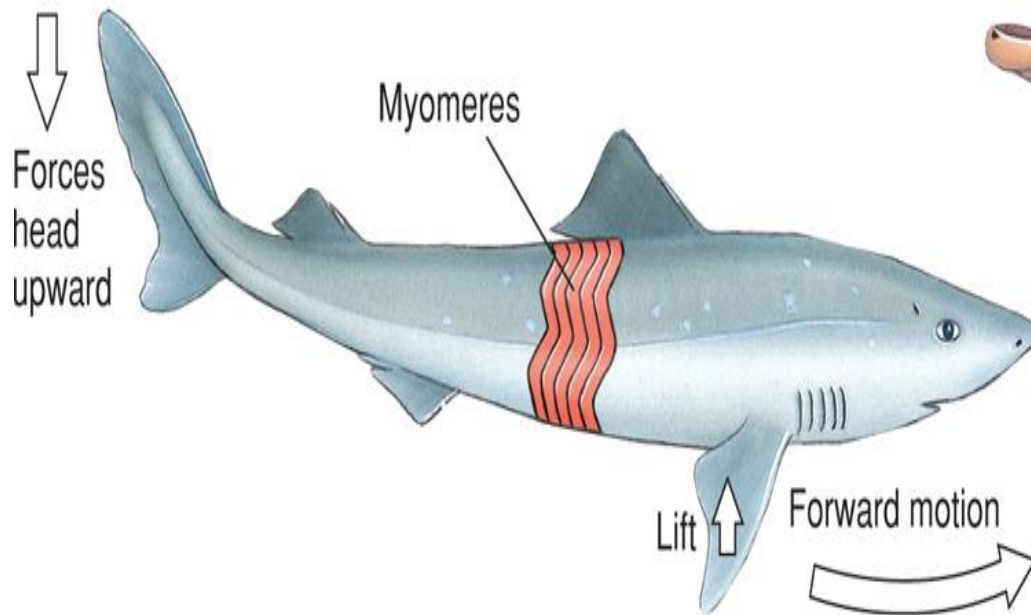


3. Locomotion

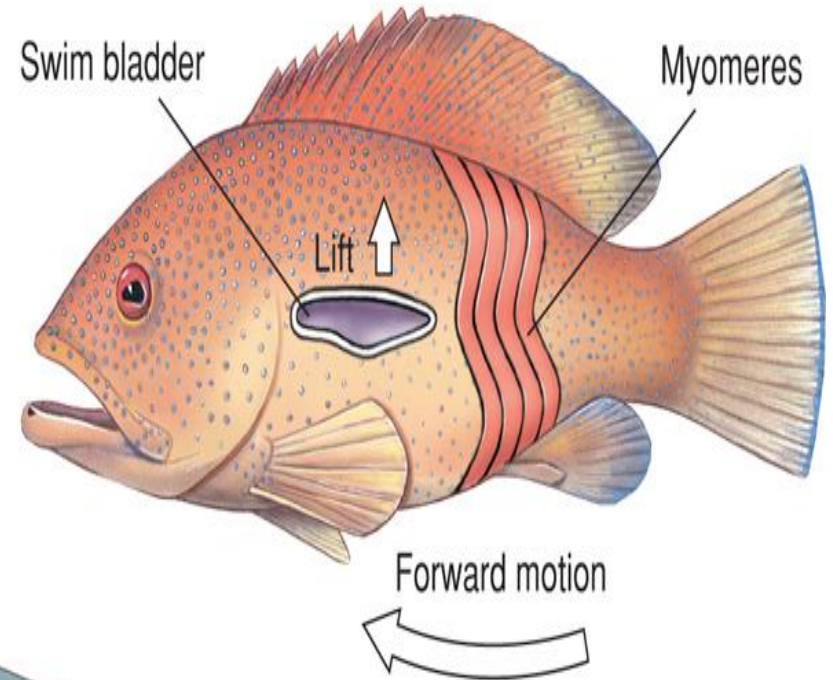
- Fish exhibit an “s-shaped” swimming pattern
- Depending on the type of fish, different fins may be used primarily for the forward movement



- Bands of muscle along the body called **myomeres** (肌节) drive swimming motion

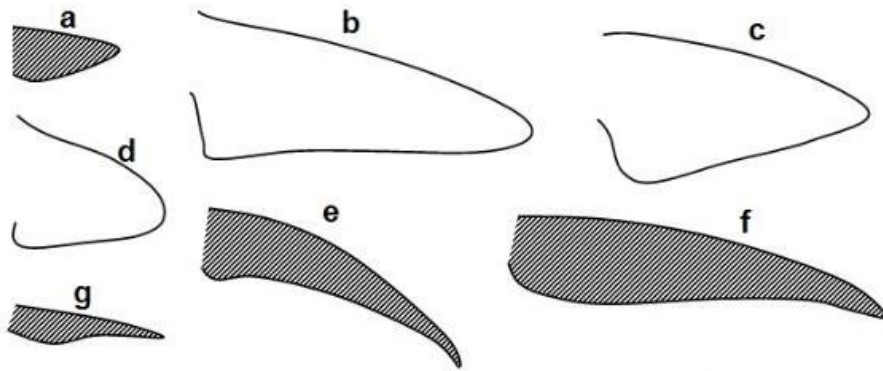


(a) Cartilaginous fishes



(b) Bony fishes

- In sharks, swim bladder is absent (**a large lipid-rich liver** to help in buoyancy). Sharks tend to sink when not in motion and there is no lift from the swim bladder
- While swimming, sharks are aided by the “lift” provided by the position and stiffness of the pectoral fins, as well as the tail

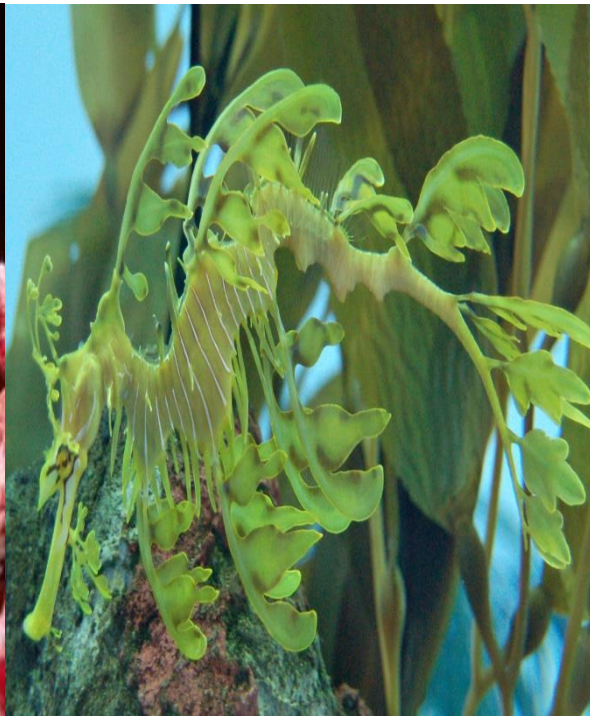


The lifting pectoral foils of different fishes (teleosts striped). (a) Mackerel (*Scomber*); (b) *Carcharhinus longimanus*; (c) basking shark (*Cetorhinus*); (d) dogfish (*Scyliorhinus*); (e) carangid (*Trachurus*); (f) longfin tuna (*Thunnus alalunga*); (g) swordfish (*Xiphias*).

a. 鲭鱼; b. 纺锤真鲨; c. 姥鲨; d. 狗头鲨; e. 鲹鱼; f. 金枪鱼(鲔鱼); g. 剑鱼

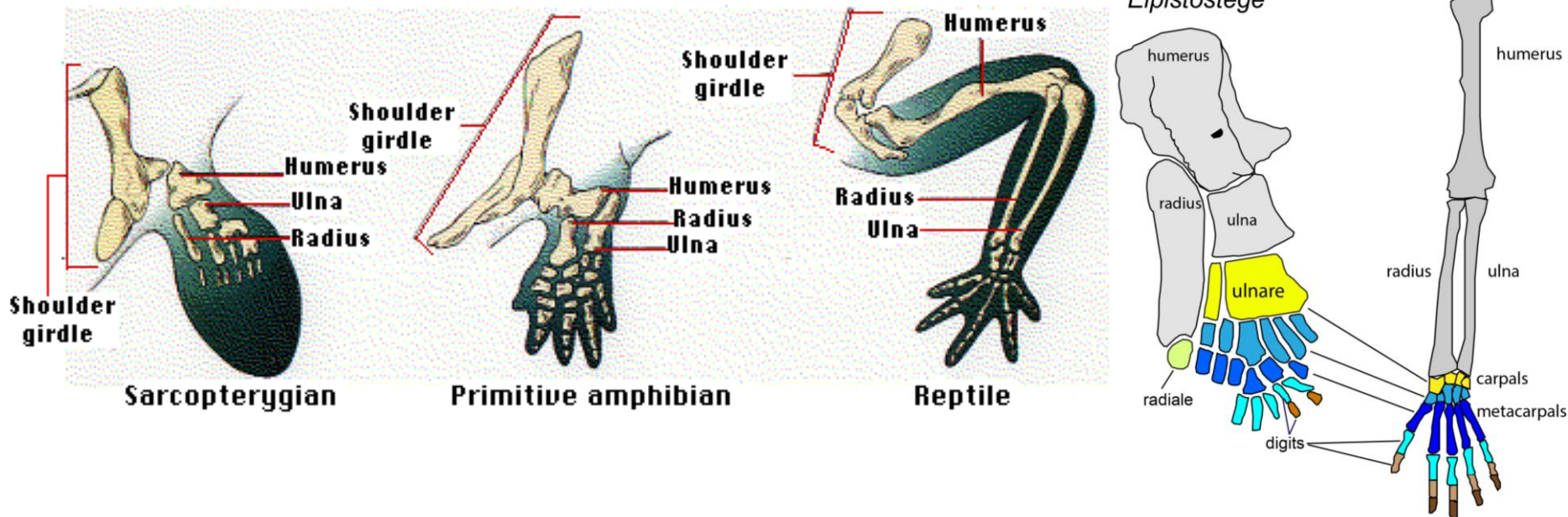
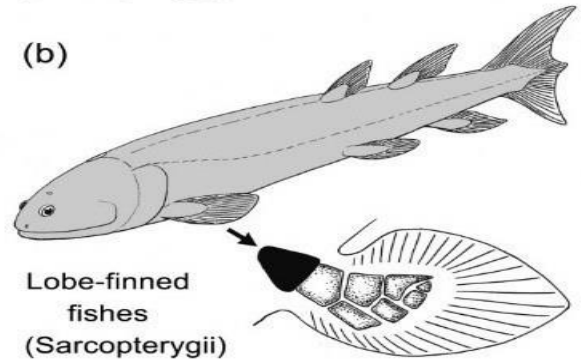
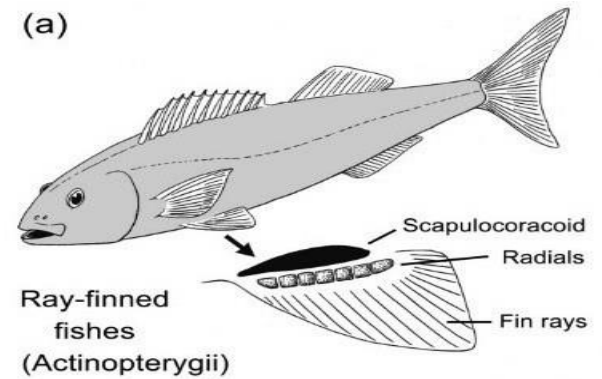


- In **bony fish**, swim bladder was applied for buoyancy, pectoral fins are not needed for lift and normally not stiff in construction. The pectoral fins in many **bony fish** are flexible
- In some slower-swimming species, forward movement is mainly provided primarily by the pectoral fins
- In other species, all the fins may be flexible and highly modified for camouflage (example: sea horses and sea dragons), means that the fins will not allow for significant forward movement



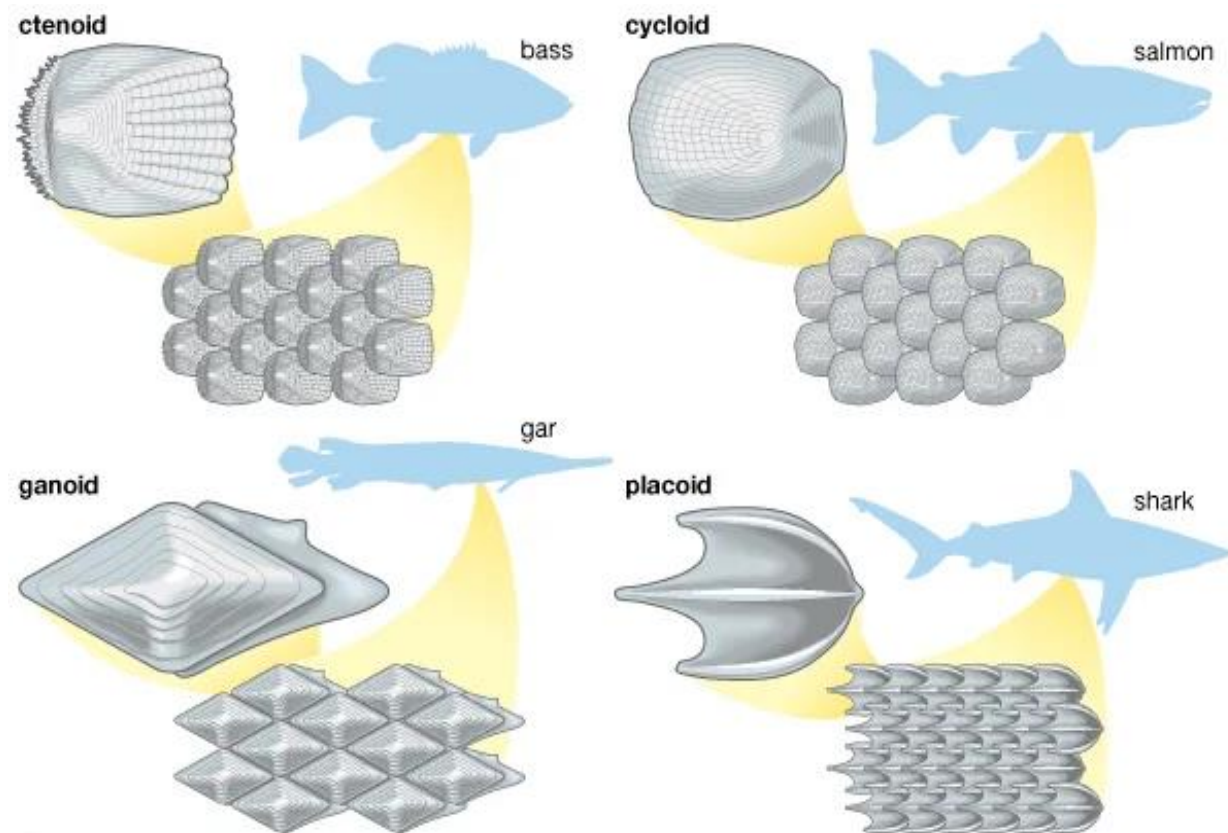
4. Fins of fish:

- **Ray-finned fishes:** Webbed spines, comprise 99% of all fish species
- **Lobe-finned fishes:** Sturdy bone structure and stronger pectoral muscles, foreshadow the evolutionary development of all tetrapods on land



5. Scales of fish

- Body of majority of fish is covered with scales
- Small scales appear on the body of the young fish when it is on the transition from the stage of the late larvae to the stage early fry
- Number of scales does not change, but their size increases with age
- There are the following types of scales: ctenoid (栉鳞), cycloid (圆鳞), ganoid (硬鳞), placoid (盾鳞)



6. Fish Gills

- The construction of the gill is the same in all fish
- **Gill arch (鳃弓)** supports the entire structure, **gill rakers (鳃耙)** are on the forward surface of the gill arch and **gill filaments (鳃丝)** trail behind the gill arch
- Exchange of oxygen and carbon dioxide takes place on these surfaces
- **Count-current flow system**

